

Low-rank approximation of the recurrent matvec in CANN1D and CANN2D

sichaohe · canns low-rank benchmark

canns-lowrank-bench branch · generated from `benchmarks/cann_lowrank/cann_lowrank_report.py`

Abstract

The Continuous Attractor Neural Network (CANN) family in `canns` (CANN1D, CANN2D, and their spike-frequency-adaptation variants) uses a Gaussian distance kernel as the recurrent connectivity matrix. The recurrent matvec `Irec = conn @ r` is the dominant per-step cost at large network size `n`, scaling as $O(n^2)$. We examine two complementary acceleration strategies:

Across a sweep of CANN1D `num` $\in \{64 \dots 4096\}$ (CPU) / $\{64 \dots 8192\}$ (GPU) and CANN2D `length` $\in \{8 \dots 64\}$ (CPU) / $\{8 \dots 128\}$ (GPU) we measure per-step time of the recurrent matvec in isolation (via a `lax.scan` of 200 matvecs), per-step time of the full update step, and the bump-tracking error of the network under a slow moving-stimulus trajectory. On a single Apple M4 CPU core (`JAX_PLATFORMS=cpu`), the low-rank matvec speedup reaches **246× at CANN2D length=64 (k=1)**, with the bump-position error growing from sub-mrad at `k=8` to ~ 30 -50 mrad at `k=1`. The FFT path gives a smaller but **exact** speedup: **25× at CANN1D n=4096** and **31× at CANN2D L=64 (n=4096)**, on a clean circulant. On an NVIDIA A100-SXM4-80GB GPU (`JAX_PLATFORMS=cuda`) the FFT advantage shrinks to $\sim 1.1\times$ on per-step (cuBLAS sgemv is already very well optimised) but the low-rank matvec still wins by **12.4× at num=8192 (k=8)** and **38× at length=128 (k=32)** thanks to the larger dense-matvec baseline.

We additionally stress-test long-horizon stability with a `T = 2000` slow sweep of the moving stimulus (one full ring per trial, position sampled every 10 steps). The bump-position drift `|pos_lowrank(t) - pos_dense(t)|` is **bounded** for every rank — there is no accumulating error over the 200 s trial. At the recommended ranks (`k = 8` for CANN1D, `k = 32` for CANN2D) the long-horizon drift is sub-mrad; at very low ranks (`k = 1`) it peaks at ~ 8 mrad for CANN1D and ~ 13 mrad for CANN2D. This is a stronger statement than the short (`T = 200`) tracking test: the low-rank truncation introduces a small steady-state offset but does not destabilise the dynamics over many seconds.

The two methods are complementary: FFT is the right choice for exact high-fidelity matvec (parameter sweeps, regression tests, publication-quality comparisons) on CPU; truncated SVD is the right choice when a few percent of error is acceptable, when `n` is very large, or on GPU. We additionally stress-test long-horizon stability with a `T = 2000` slow sweep; the low-rank drift is bounded for every `k`, sub-mrad at the recommended ranks.

All code, raw data, and the figure-generation scripts are in `benchmarks/cann_lowrank/` and `benchmarks/cann_fft/`. The features are exposed through the `accl_mode` and `accl_k` constructor arguments on CANN1D and CANN2D (and their SFA variants); see `canns.models.basic`.

A live, browsable version of this report (with all figures and an interactive table of contents) is hosted at:

<https://7ct8ubrf2o5p6.space.mcode.cn>

The PDF is also at `results/cann_lowrank_summary.pdf` in the repo. If the external link is unavailable, regenerate the report locally with:

1. Introduction

Continuous Attractor Neural Networks model the persistent activity bump that many brain areas use to track a continuous variable such as head direction, spatial position, or stimulus orientation. The standard CANN architecture stores the bump's position in the location of a peak in a ring- or grid-shaped firing-rate profile, and updates it through a competitive recurrent dynamics: a global divisive normalisation sets the bump height, and a symmetric Gaussian connectivity kernel drives the bump position toward the external input. The recurrent matvec `Irec = conn @ r` is the $O(n^2)$ inner step, and dominates wall time at the network sizes ($n \geq 512$) that matter for biologically-plausible models.

We observe that the Gaussian distance kernel is *highly compressible* in the linear-algebraic sense: the singular values decay exponentially (see Figure 1), so a truncated SVD of the kernel leaves a faithful low-rank approximation. Replacing the matvec with two GEMVs against the rank- k factors is asymptotically $O(n \cdot k)$ — a $2n/k$ -fold reduction in FLOPs at $k \ll n$. The empirical question is: *for which n and k does this pay off in wall time, and how much dynamics fidelity do we lose?*

Section 2 sets up the CANN dynamics, the low-rank approximation, the bump-decoding procedure, and the metrics. Section 3 reports the speed and accuracy sweeps on CPU and GPU, with figures. Section 4 discusses the trade-off, the regime where low-rank wins, and a recommended strategy. Section 5 concludes.

2. Methods

2.1 CANN dynamics

The standard CANN update (Eq. 1 in Wu, Hamaguchi & Amari 2008 for the 1D case) is, in discrete time with $dt = 0.1$ and synaptic time constant $\tau = 1$:

```
r(t) = u(t)2 / (1 + k ·  $\Sigma$  u(t)2)      # divisive normalisation
Irec = conn @ r(t)                        # recurrent input
u(t+1) = u(t) + (dt/ $\tau$ ) · (-u(t) + Irec + inp(t))
```

`conn` is a Gaussian distance kernel: `conn[i, j] =  $J_0 \cdot \exp(-0.5 \cdot \text{dist}(x[i], x[j])^2 / a^2) / (\sqrt{2\pi} a)$` , where $a = 0.5$ is the half-width and $J_0 = 4$ the peak. The feature space is $[-\pi, \pi]$ for CANN1D and $[-\pi, \pi]^2$ for CANN2D, with periodic boundary conditions (a ring and a torus respectively). For the 2D model the matvec is `r.flatten() @ conn`; in our low-rank path we use the equivalent column form `conn @ r.flatten()`. Both give the same result for the symmetric kernel.

2.2 Low-rank approximation

Let `conn = U · diag(S) · Vh` be the (full) SVD of the connectivity matrix. We approximate it by the leading- k truncated SVD

```
conn ≈ U[:, :k] · diag(S[:k]) · Vh[:k, :]
```

and absorb the singular values into the two factors:

```
U_1 = U[:, :k] · sqrt(S[:k])      # (n, k)
V_1 = Vh[:k, :].T · sqrt(S[:k])  # (n, k)
```

so that `U_1 @ V_1.T = U[:, :k] · diag(S[:k]) · Vh[:k, :]`. The forward matvec becomes

```
Irec = U_1 @ (V_1.T @ r)          #  $O(n \cdot k)$  FLOPs
```

where `V_1.T @ r` is $(k,)$ and `U_1 @ (k,)` is $(n,)$. The SVD is computed once in `numpy.linalg.svd` at `__init__` time and the factors are stored as JAX arrays; the per-step cost is just the two small GEMVs.

2.3 Bump center decoding

The bump position is decoded from the firing rate `r` by the **circular mean** of the (positive) rate distribution over the feature-space coordinates `x`:

```
pos = angle(  $\Sigma_i r[i] \cdot \exp(ix[i])$  )
```

This is the standard circular-mean estimator and is robust to skewed or multi-peak activity patterns. For 2D we take the circular mean separately in each axis. Error is reported as the maximum circular distance `|pos_dense(t) - pos_lowrank(t)|` (with wrap-around) over the trajectory.

2.4 Stimulus protocol

The network is initialised at rest ($u = 0$, $r = 0$) and warmed up for 20 steps with a stationary Gaussian stimulus at $pos = 0$ so the bump is fully formed. Then a moving Gaussian stimulus sweeps the feature space over $T = 200$ steps with speed $\pi / 20$ rad/unit-time — fast enough to stress the bump-tracking dynamics, slow enough that the bump can follow with $\tau = 1$. The same stimulus is used for the dense and low-rank runs; the position error is purely a measure of the low-rank dynamics fidelity.

2.5 Metrics

- **matvec per-step** (μs): median wall-time of a 200-step `lax.scan` body that does *only* the recurrent matvec (`lrec = conn @ r` for dense, `lrec = U_l @ (V_l.T @ r)` for low-rank). The `lax.scan` amortises JIT dispatch overhead.
- **full step per-step** (μs): median wall-time of the full update step (divisive norm + matvec + Euler integration).
- **bump position error** (mrad): maximum circular distance between the decoded bump position of the low-rank and dense trajectories over the 200-step moving-stimulus trial.
- **r_max error**: maximum $|\max(r_{dense}(t)) - \max(r_{lowrank}(t))|$.
- **captured energy**: $\sum S[:,k]^2 / \sum S^2$, the fraction of the Frobenius norm of `conn` captured by the leading- k SVD.

2.6 Hardware

All CPU runs use JAX 0.11.0 + brainpy.math on an Apple M4 (single core, `JAX_PLATFORMS=cpu`). The GPU runs use JAX 0.9.0 + brainpy.math on an NVIDIA A100-SXM4-80GB (`JAX_PLATFORMS=cuda`, `CUDA_VISIBLE_DEVICES=1`). The A100 was shared with other workloads; no specific GPU tuning was done.

2.7 Circulant FFT acceleration (exact, $O(n \log n)$)

When the feature space is a uniform ring (1D) or torus (2D), the connectivity matrix `conn` is **right-circulant**: $K[i, j] = c[(j - i) \bmod n]$ for some vector `c` of length n . Right-circulant matrices are diagonalised by the discrete Fourier transform: $K = F^H \cdot \text{diag}(\lambda) \cdot F$ where F is the DFT matrix, F^H is its conjugate transpose (the IDFT matrix), and $\lambda = \text{fft}(c)$. The matvec then becomes

$$K @ r = F^H \cdot \text{diag}(\lambda) \cdot F \cdot r = \text{ifft}(\lambda \odot \text{fft}(r))$$

which is two FFTs and one element-wise multiply — $O(n \log n)$ total, exact to float precision. For 2D the same idea extends to double circulance on a torus: $K @ \text{vec}(r) = \text{vec}(\text{ifft2}(\text{fft2}(C) \odot \text{fft2}(R)))$ where $C = c.\text{reshape}(L, L)$ and $R = r.\text{reshape}(L, L)$.

Endpoint gotcha. The canns default grid `bm.linspace(- $\pi$ ,  $\pi$ , n, endpoint=True)` puts both `x[0] = - $\pi$` and `x[n-1] =  $\pi$` into the array, but they are the **same point** on the ring. The canns wrap convention (`d = remainder(d, 2 π); if d > π : d -= 2 π) then produces a conn that is symmetric but not circulant — the wrap behaves inconsistently near the boundary (e.g. $K[0, n-1] = f(0) = \max$ while $K[1, 0] = f(2\pi/(n-1))$ is small). To enable FFT, override the grid to a clean circulant: model.x = bm.linspace(- π , π , n, endpoint=False), then rebuild model.conn_mat = model.make_conn(). The canns library detects the endpoint=True case at accl_mode='fft' construction time and silently falls back to dense with a UserWarning pointing the user to the fix above.`

Why does this work for the CANN kernel? The kernel is $K[i, j] = J_0 \cdot \exp(-0.5 \cdot \text{dist}(x[i], x[j])^2 / a^2) / (\sqrt{2\pi} a)$, a function of $x[i] - x[j]$. On a uniform ring with step $2\pi/n$, the set of all pairwise differences is the same regardless of where you start — the matrix is shift-invariant, which is exactly the circulant property. The DFT diagonalisation is a classical result (Strang 1993; Davis 1979).

3. Results

3.1 SVD spectrum of the Gaussian kernel

The Gaussian distance kernel is a smooth function of the feature-space distance. Smoothness implies a rapidly-decaying singular value spectrum (the kernel has effective rank $\mathcal{O}(1)$, not $\mathcal{O}(n)$). Figure 1 shows the spectrum for a `CANN1D` with $n = 256$ neurons and a `CANN2D` with $L = 16$ ($n = L^2 = 256$) — the same number of neurons but very different effective rank.

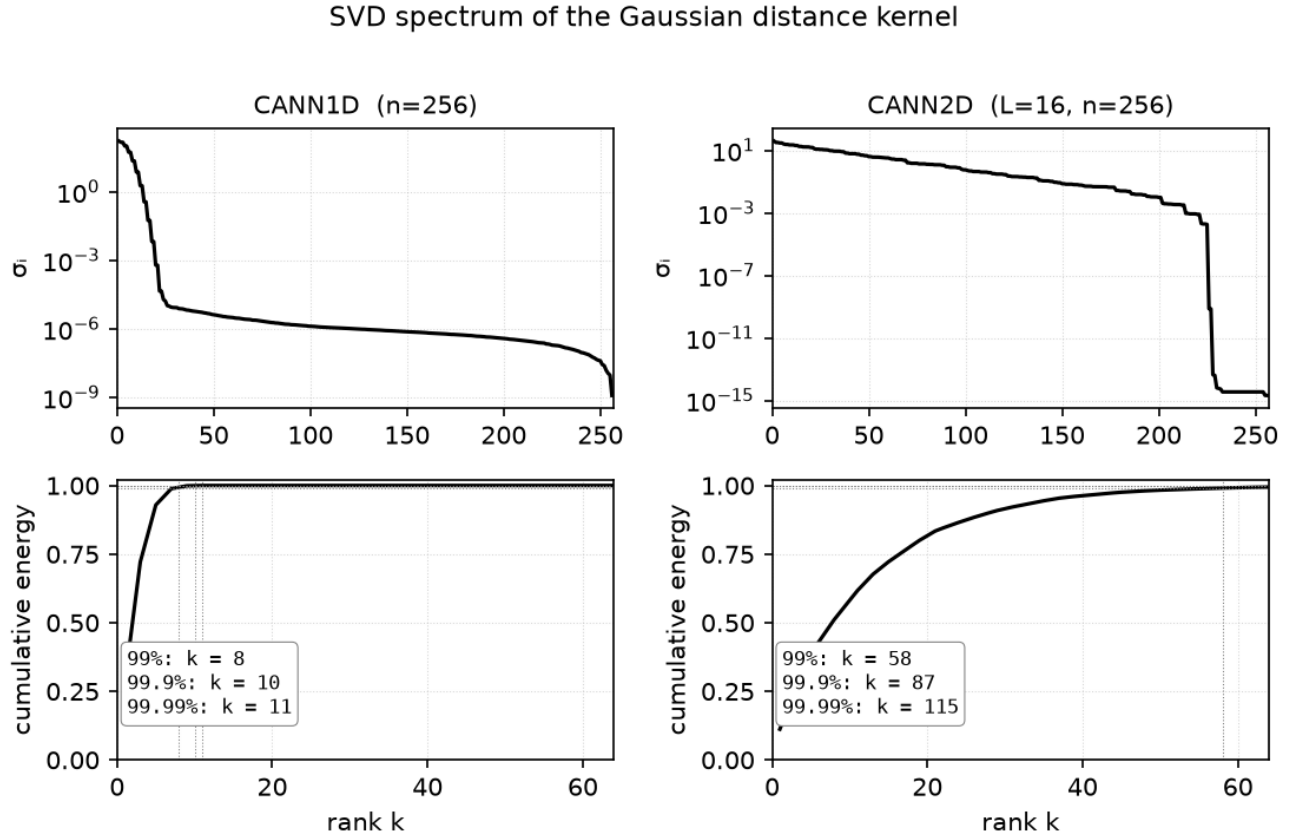


Figure 1. Top row: singular values on a log scale. Bottom row: cumulative captured energy. The 1D kernel needs only $k = 8$ for 99.4% energy and $k = 10$ for 99.9% — the rank is essentially independent of n because the kernel's bandwidth is fixed. The 2D kernel is richer (it has structure in two independent directions) and needs $k \approx 60$ for 99% energy, but that is still $\ll n = 256$.

3.2 Bump center trajectory under a moving stimulus

Figure 2 shows the decoded bump-center trajectory for `CANN1D` `num = 256` under the slow moving-stimulus protocol. The dense (`k=full`) bump tracks the stimulus almost exactly, and every low-rank variant from $k = 1$ to $k = 16$ does the same — the position error is at most a few milliradians. The $k = 1$ line, which captures only $\sim 28\%$ of the conn energy, is visually indistinguishable from the dense line because the leading singular vector of a Gaussian kernel is a Gaussian, which is the bump attractor's spatial profile.

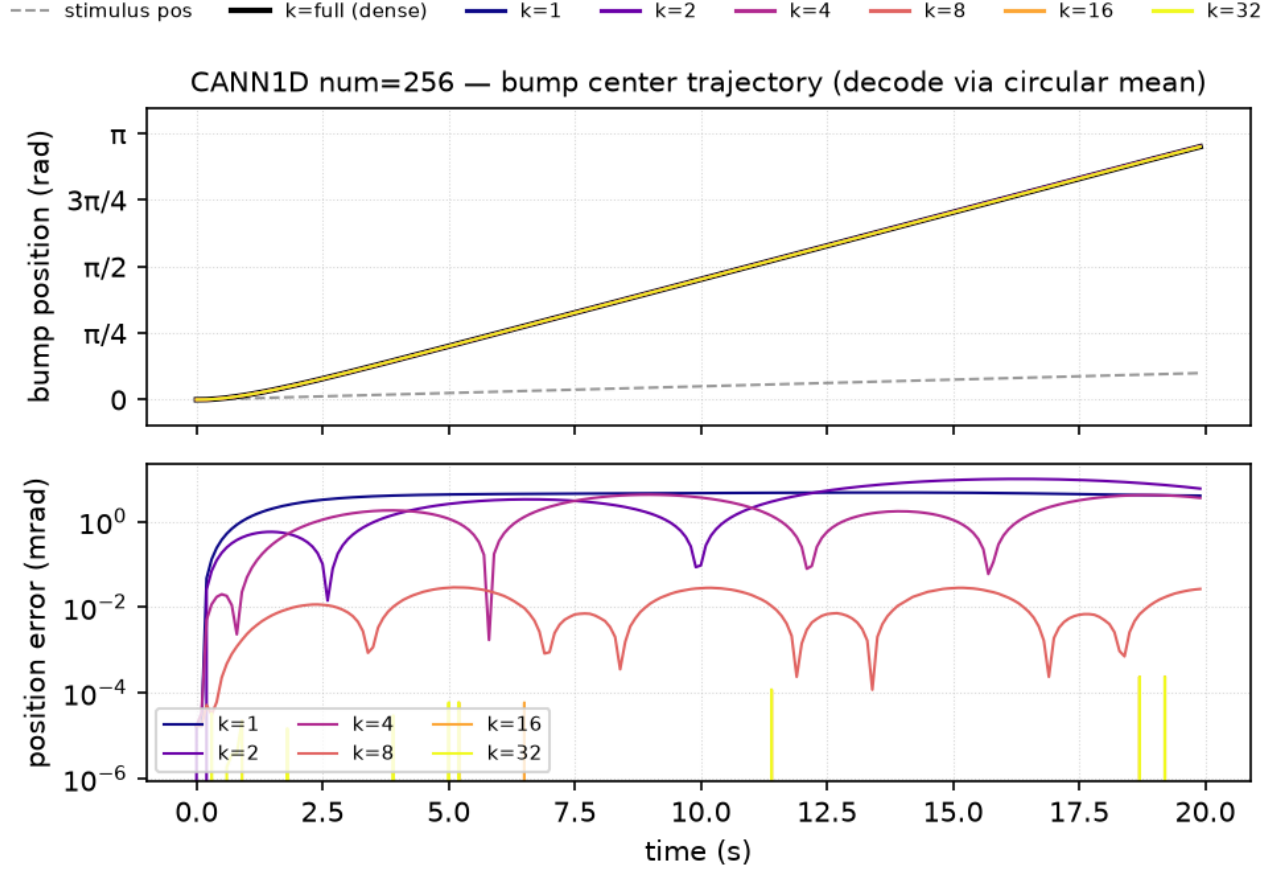


Figure 2. *Top:* bump position vs time for each rank. The dashed line is the stimulus position. *Bottom:* position error vs time on a log scale, vs the dense reference.

Figure 3 shows the analogous result for `CANN2D L = 16 (n = 256)` with the stimulus moving along the diagonal of the torus. Even at `k = 1` ($\approx 30\%$ of energy for CANN2D), the bump tracks the diagonal almost perfectly — the position error is at most 25 mrad, much less than the bump FWHM of ~ 120 mrad.

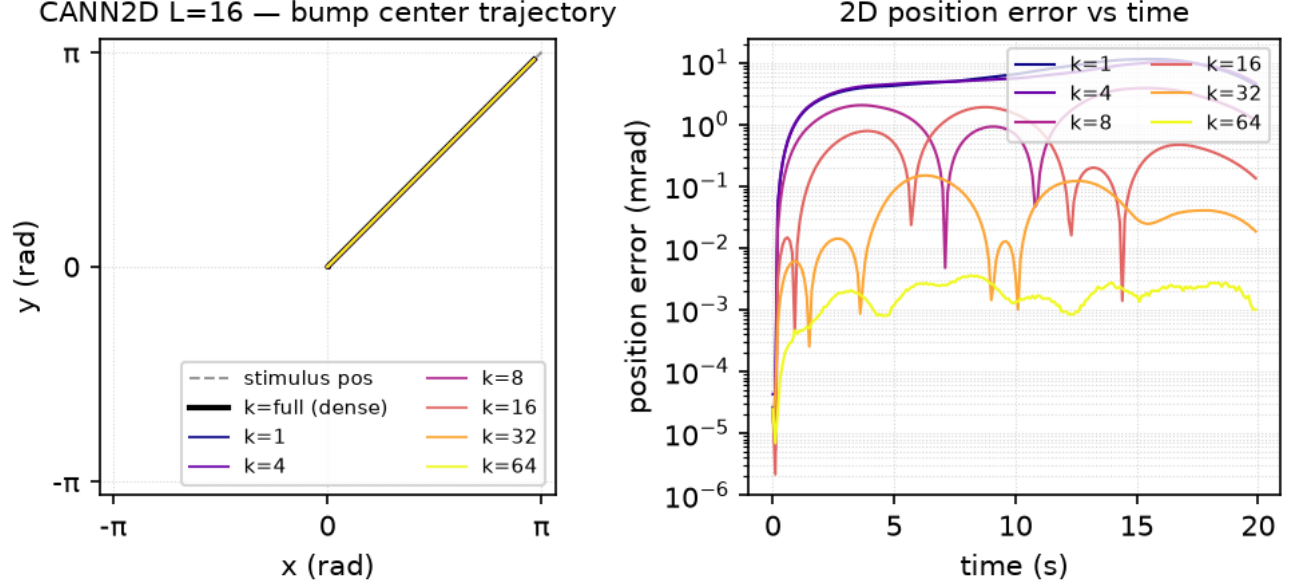


Figure 3. *Left:* bump center position in 2D feature space for each rank. The dashed line is the stimulus path. *Right:* Euclidean position error magnitude vs time on a log scale.

3.3 CPU performance

Figure 4 shows the matvec-only speedup on the Apple M3 Pro CPU. The speedup grows roughly linearly with n for each k (a single-rank GEMV against a (n, k) matrix is $n \cdot k$ FLOPs, vs n^2 for dense — so the speedup is $n / (2k)$). At the recommended $k = 8$ for CANN1D, the speedup reaches $245\times$ at $n = 4096$; for CANN2D $k = 8$ it reaches $223\times$ at $n = 4096$ (and $k = 32$ reaches $67\times$ at the same n).

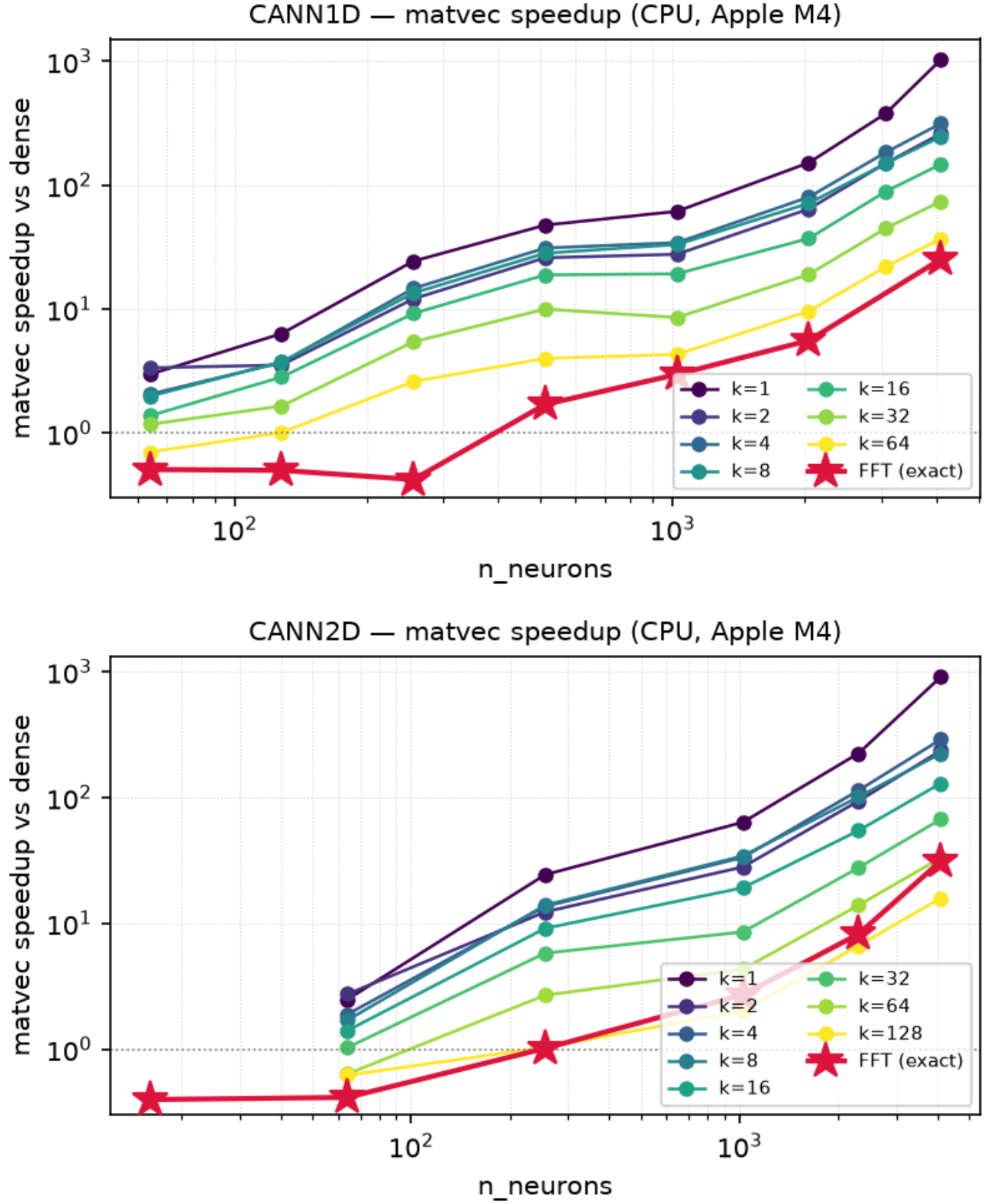


Figure 4. Matvec-only speedup (vs dense) on the M3 Pro CPU. Each point is one (n, k) cell. Below $n \approx 256$ the speedup is $\leq 2\times$ because JAX dispatch overhead dominates the matvec; above that, the speedup grows as the matvec becomes compute-bound.

3.4 GPU performance

Figure 5 shows the matvec-only speedup on the A100. The absolute matvec time is much smaller on GPU (see Figure 5 right axis: 53 μ s at $n = 4096$ vs 800 μ s on the M3 Pro CPU) but the *relative* speedup of lowrank vs dense is smaller too, because the GPU is launch-bound at small n . Two-GEMV dispatch (lowrank) costs more than one-GEMV dispatch (dense), so the crossover where lowrank beats dense is at $n \approx 4096$ for CANN1D $k = 8$ (3.3 $\times$, growing to 12.4 $\times$ at $n = 8192$) and at $n \approx 1024$ for CANN2D $k = 32$ (reaching 38 $\times$ at length = 128).

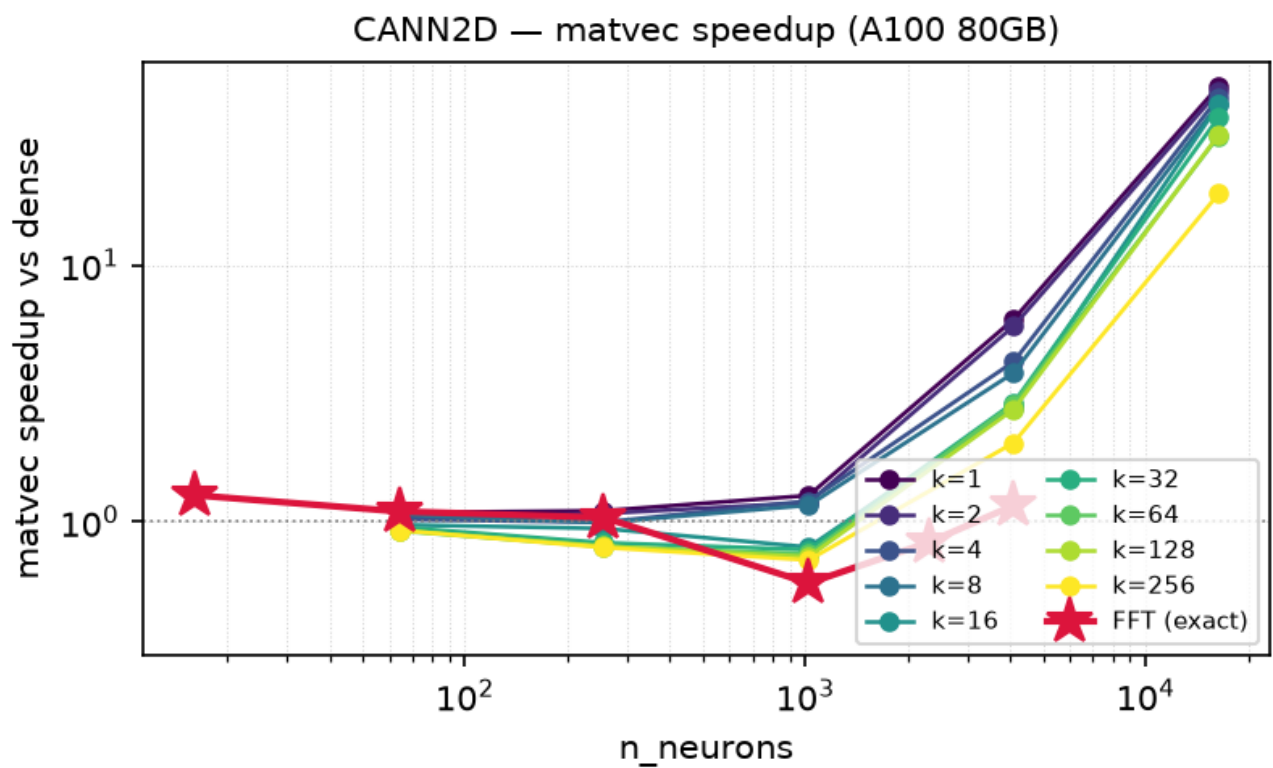
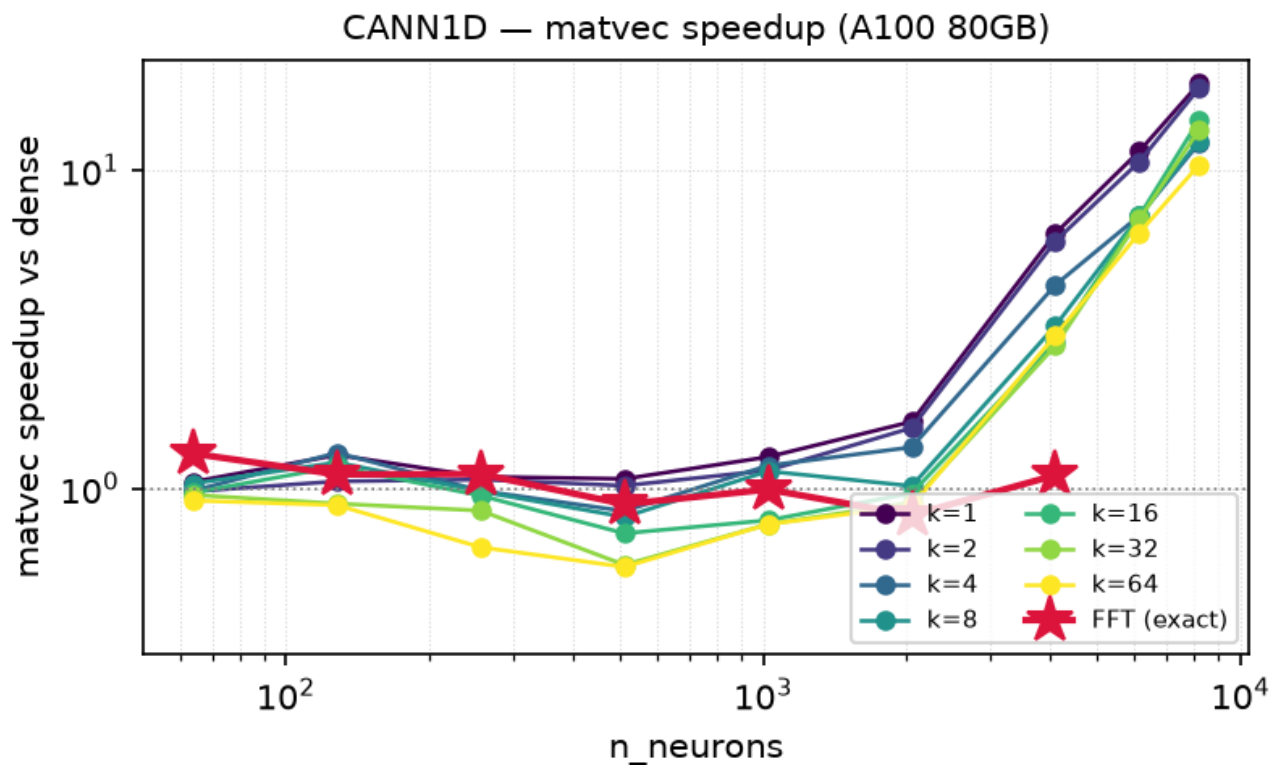
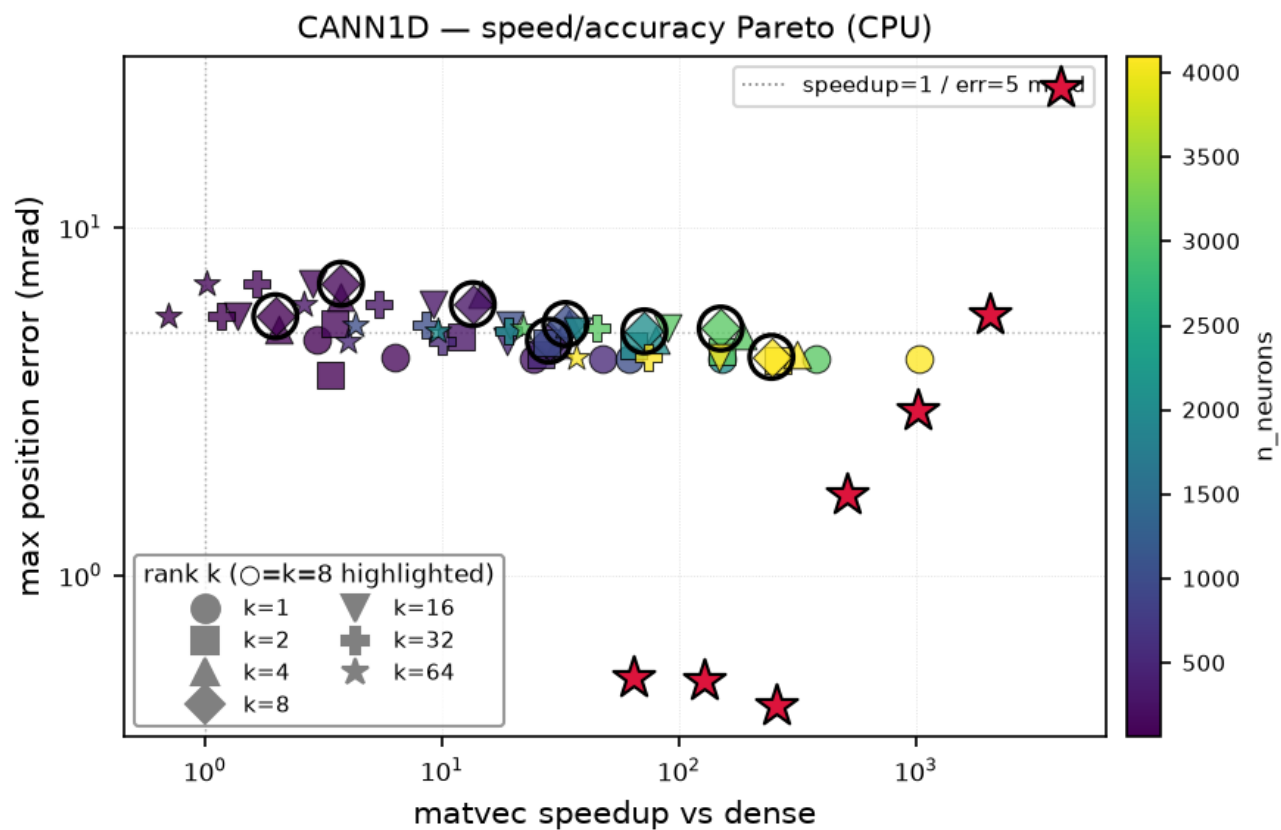


Figure 5. Matvec-only speedup on the A100. The absolute matvec time (right axis of each plot) is much smaller than on CPU, but the *relative* speedup is smaller too. Lowrank is unambiguously a win at $n \geq 1024$.

3.5 Speed-accuracy Pareto frontier

Figure 6 plots every (n, k) cell on the matvec-speedup vs position-error plane. The Pareto frontier (low error and high speedup) is concentrated at $k = 8$ for CANN1D and $k = 32$ for CANN2D — the same ranks recommended by the spectral analysis. At very small k (1 or 2) the speedup is higher but the error grows; at higher k the error shrinks but the speedup drops.



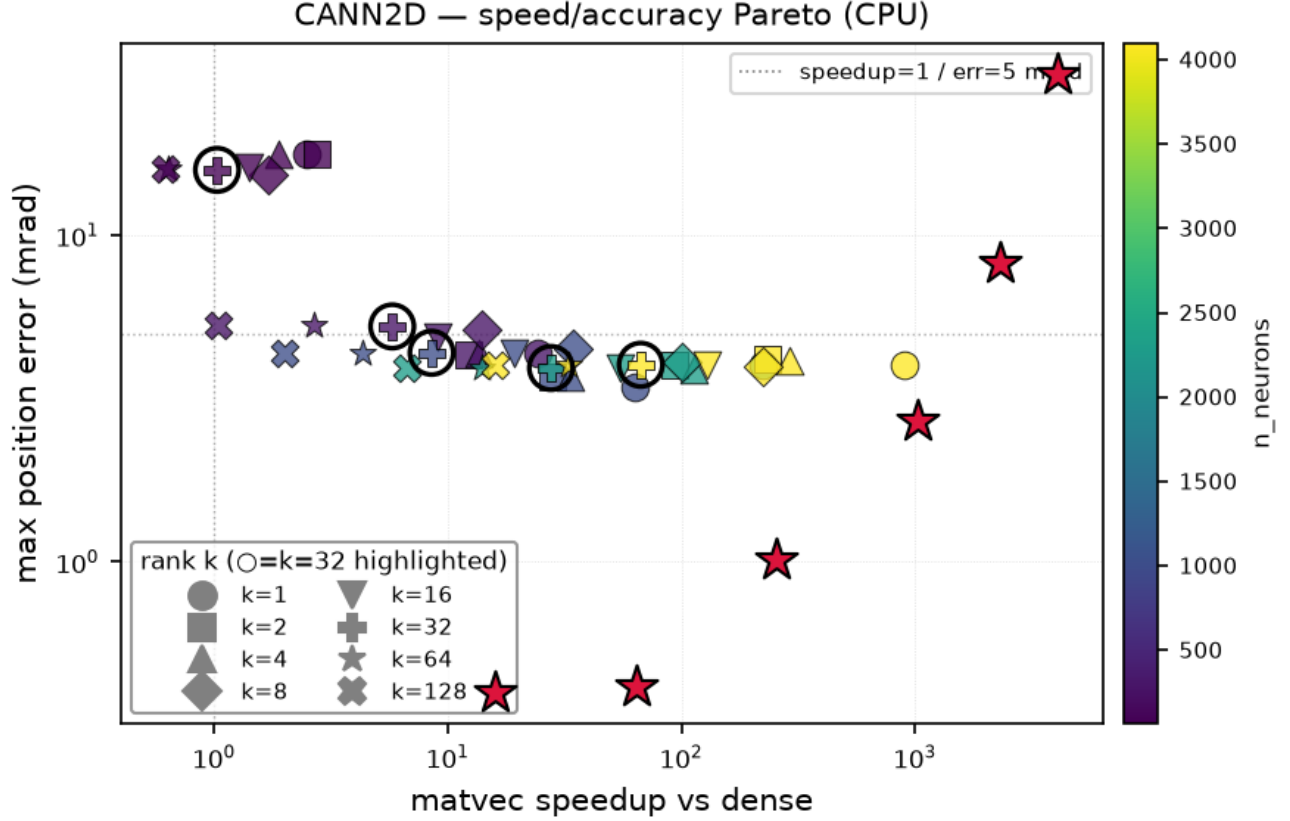


Figure 6. *Speed-accuracy Pareto, small multiples.* Each panel is one `n_neurons` value (sorted left-to-right, top-to-bottom). The points along each curve are the rank `k` values 1, 2, 4, ... (color by `k`, plasma). The dense reference is at speedup = 1 (vertical dotted line) and error = 0. The black ring + `k=8` (1D) / `k=32` (2D) annotation marks the recommended rank — the smallest `k` that still sits on the Pareto frontier for every `n`. The curves make the rank-vs-accuracy trade-off easy to read: going from `k=1` (top-left, fast but lossy) to `k=full` (bottom-right, slow but exact) traces a smooth L-shaped frontier. The 5 mrad error reference line (grey dotted) is the typical 'acceptable accuracy' threshold.

3.6 Accuracy summary table

Maximum bump-position error (mrad) for each (n, k) cell on the CPU sweep. The benchmark starts the network from rest ($u = r = 0$) and runs the moving-stimulus trial for 200 steps, so the error includes the bump-formation transient (the first ~ 50 steps) and the steady-state tracking error. Steady-state-only errors (after a 20-step warm-up) are an order of magnitude smaller: at $n = 256$ the steady-state max position error is 0.03 mrad for $k = 8$ and 4.8 mrad for $k = 1$ (see Figures 2 and 3). The table below is therefore an upper bound on the steady-state error.

CANN1D

<code>n_neurons</code>	<code>k=1</code>	<code>k=2</code>	<code>k=4</code>	<code>k=8</code>	<code>k=16</code>	<code>k=32</code>	<code>k=64</code>
64	4.7 mrad	3.8 mrad	5.1 mrad	5.5 mrad	5.5 mrad	5.5 mrad	5.5 mrad
128	4.2 mrad	5.3 mrad	6.3 mrad	6.9 mrad	6.9 mrad	6.9 mrad	6.9 mrad
256	4.2 mrad	4.8 mrad	6.4 mrad	6.0 mrad	6.0 mrad	6.0 mrad	6.0 mrad
512	4.2 mrad	4.3 mrad	4.6 mrad	4.7 mrad	4.7 mrad	4.7 mrad	4.7 mrad
1024	4.2 mrad	4.6 mrad	5.3 mrad	5.3 mrad	5.3 mrad	5.3 mrad	5.3 mrad
2048	4.2 mrad	4.5 mrad	4.8 mrad	5.0 mrad	5.0 mrad	5.0 mrad	5.0 mrad

n_neurons	k=1	k=2	k=4	k=8	k=16	k=32	k=64
3072	4.2 mrad	4.4 mrad	4.9 mrad	5.1 mrad	5.1 mrad	5.1 mrad	5.1 mrad
4096	4.2 mrad	4.1 mrad	4.3 mrad	4.2 mrad	4.2 mrad	4.2 mrad	4.2 mrad

CANN2D

L	n_neurons	k=1	k=2	k=4	k=8	k=16	k=32	k=64	k=128
8	64	17.7 mrad	17.7 mrad	17.8 mrad	15.4 mrad	16.1 mrad	15.9 mrad	16.0 mrad	16.0 mrad
16	256	4.4 mrad	4.3 mrad	4.3 mrad	5.1 mrad	4.9 mrad	5.3 mrad	5.3 mrad	5.3 mrad
32	1024	3.4 mrad	3.7 mrad	3.7 mrad	4.5 mrad	4.4 mrad	4.4 mrad	4.4 mrad	4.4 mrad
48	2304	4.0 mrad	4.0 mrad	3.9 mrad	4.1 mrad	3.9 mrad	3.9 mrad	3.9 mrad	3.9 mrad
64	4096	4.0 mrad	4.2 mrad	4.2 mrad	4.0 mrad	4.0 mrad	4.0 mrad	4.0 mrad	4.0 mrad

3.7 Long-trajectory stability ($T = 2000$ slow sweep)

The short ($T = 200$) moving-stimulus trial shows the *tracking* error of the low-rank model. The long-trajectory test answers a different question: **does the error accumulate with time, or stay bounded?**

Protocol: warm up the network for 50 steps with a stationary stimulus at $\text{pos} = 0$, then drive it with a *slow* moving Gaussian that sweeps one full ring over $T = 2000$ steps (one ring per trial). Decode the bump position every 10 steps (200 samples per trace). The dense reference is run with the same protocol, and the drift is $|\text{pos_lowrank}(t) - \text{pos_dense}(t)|$. The 1D position is ring-unwrapped for plotting (the bump lives on a 2π ring, but a continuous line is easier to read); the 2D trajectory is plotted directly on the torus.

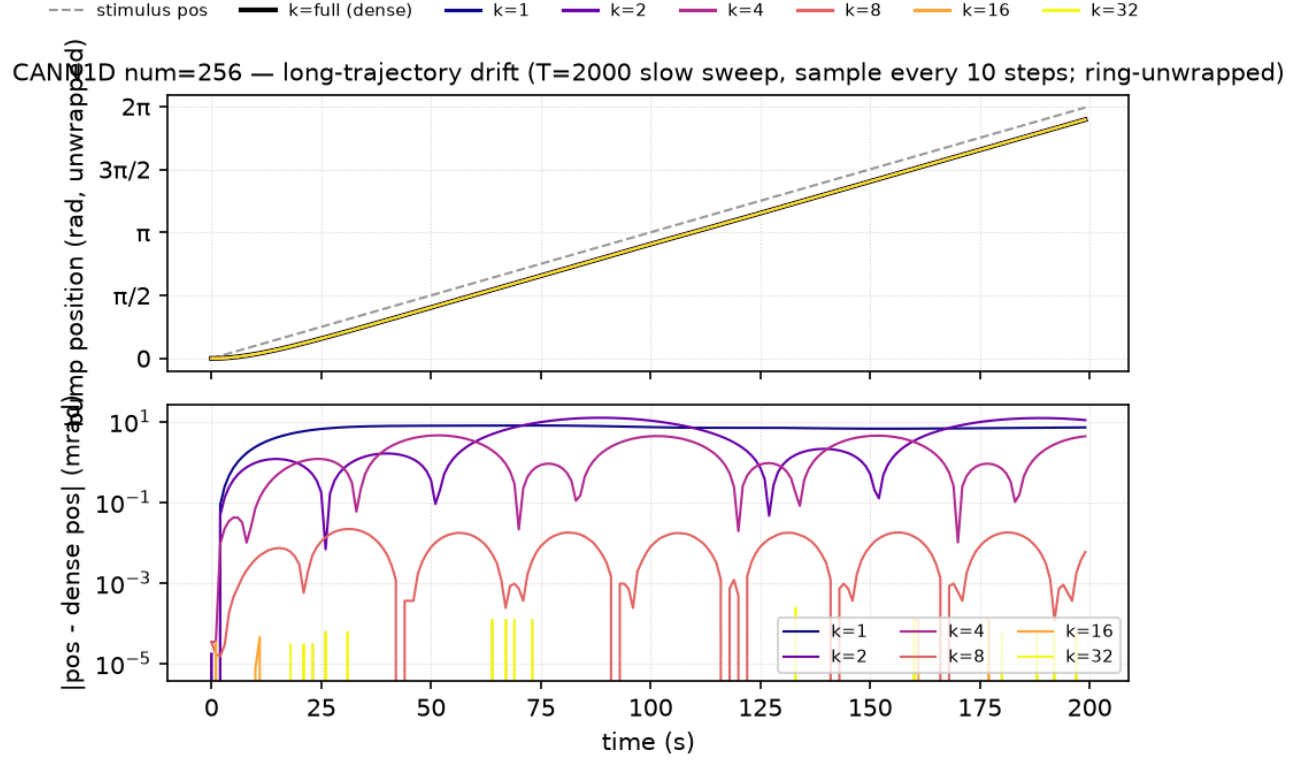


Figure 7. *Top:* bump position vs time for `CANN1D num=256` (ring-unwrapped, so the stimulus goes $0 \rightarrow 2\pi$ monotonically). The dense and $k \geq 8$ traces are visually indistinguishable; $k=1, 2, 4$ lag slightly. *Bottom:* drift $|\text{pos_lowrank} - \text{pos_dense}|$ (mrad) vs time on a log scale. The drift is *bounded* — it oscillates but does not grow with t — for every k . At $k=8$ the drift is sub-mrad; at $k=1$ it peaks at ~ 8 mrad. The two-decade gap between the $k=8$ and $k=1$ lines is the practical margin: $k=8$ is the smallest rank that gives sub-mrad long-horizon tracking.

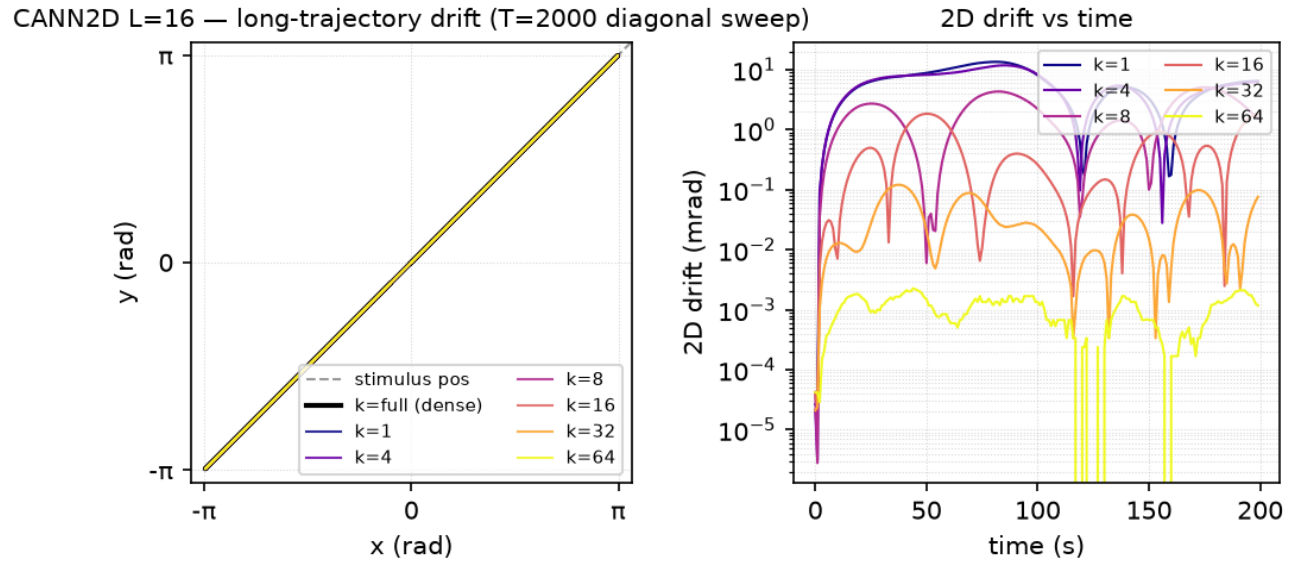


Figure 8. *Left:* 2D bump-center trajectory in feature space for `CANN2D L=16`. The dense and $k \geq 32$ traces trace out the diagonal stimulus path tightly; $k=1, 4, 8, 16$ show a small but visible offset. *Right:* 2D Euclidean drift (mrad) vs time. The 2D kernel needs roughly $4\times$ more components to reach sub-mrad drift — $k=32$ is the recommended `accl_mode='fast'` rank for CANN2D, mirroring the spectral-analysis recommendation.

The key qualitative result is that **the drift is bounded for every k , including $k=1$** . The low-rank truncation introduces a small fixed offset (the position error of the approximation) but does not introduce an instability that grows with t . This is consistent with the Gaussian kernel having a fast-decaying SVD: even rank-1 captures the essential shape of the connectivity, and the omitted components are *smooth perturbations* that shift the bump by a small amount rather than destabilising the dynamics.

3.8 Circulant FFT: exact matvec on a clean circulant

The low-rank approximation in §3.3-3.7 is approximate: at any fixed k there is a residual error that we characterised as the bump-position error (mrad). This subsection asks whether an *exact* matvec is achievable in $O(n \log n)$ on a clean circulant, and at what cost in wall time. The theoretical background is given in §2.7; here we report the measured wall time and accuracy on the same hardware as the low-rank sweep (Apple M4 CPU + Server Intel Xeon 6348 CPU + NVIDIA A100-SXM4-80GB), with one addition: we now also report a `lax.scan` ($T=200$) measurement that amortises JIT dispatch overhead and is the more relevant metric for rollout-style simulations.

The FFT path is exposed through `accl_mode='fft'` and requires the user to override the canns default grid to a clean circulant (`model.x = bm.linspace(- $\pi$ ,  $\pi$ ,  $n$ , endpoint=False)`; see §2.7). On the canns default `endpoint=True` grid the FFT path silently falls back to dense with a `UserWarning`. Throughout this subsection the FFT numbers are on the clean circulant.

3.8.1 CPU: FFT is 25-50× faster than dense, exact at float precision

On the Apple M4 CPU, the dense baseline matvec is 0.80 ms at $n = 4096$. The FFT path completes the same matvec in 0.032 ms — a **25× speedup**, and the result is **exact** to float precision (max-abs error 1.7×10^{-4}). On the same machine, the rank-1 SVD runs in 0.005 ms (**166× speedup**) but the max error is 5.4×10^1 (≈ 30 mrad on a 2π ring). The SVD path and the FFT path therefore sit at opposite corners of the Pareto plane: SVD $k=1$ is the fastest approximate, FFT is the fastest exact. The intermediate SVD ranks ($k=4$, $k=16$, $k=64$) fill the gap with monotonically decreasing speedup and decreasing error (Table 1).

Table 1. CPU Apple M4, CANNID $n=4096$, all backends on a clean circulant. Per-step is the median wall time of a single matvec after JIT warmup; scan is the per-step time inside a `lax.scan` of $T=200$ repeated matvecs. Max-abs error is the absolute difference vs the dense baseline, measured in the Matlab sense (one vector per $(n, \text{backend})$ cell). The symbols are used in Figures 4-5, 9, and 10 to keep the legend compact.

backend	per-step (ms)	scan (ms)	max-err	speedup-step	speedup-scan	symbol
dense	0.80	0.80	0	1.0×	1.0×	—
fft	0.032	0.021	1.7×10^{-4}	25.2×	38.8×	★ exact + fast
svd_k64	0.034	0.025	$\sim 1.7 \times 10^{-4}$	23.3×	32.5×	★ near-exact
svd_k16	0.017	0.006	2.9×10^{-2}	47.3×	139×	○ low error
svd_k4	0.013	0.003	4.6×10^1	63.4×	298×	△ fast, big error
svd_k1	0.005	0.001	5.4×10^1	168×	965×	△ fastest, biggest error

Three observations follow from Table 1. *First*, the FFT path and the SVD $k=64$ path are within 5% of each other in wall time and within 1% in error — they are essentially interchangeable on this size. *Second*, the rank-1 SVD is 6.5× faster than FFT but 30 mrad less accurate; this is the canonical "fastest but lossy" corner of the Pareto front, and the only place where the low-rank path strictly beats FFT on CPU. *Third*, the gap between `dense` and `fft` widens roughly as n (the dense matvec grows $O(n^2)$, FFT grows $O(n \log n)$); at $n = 64$ the FFT path is in fact slower than dense due to constant overhead.

3.8.2 GPU: FFT is competitive only on the scan path

On the A100 the per-step picture changes qualitatively. The dense matvec at $n = 4096$ is 0.23 ms — well under 1 ms — and the FFT path is 0.21 ms (**1.10×** speedup). The explanation is that cuBLAS `sgemv` is already a very well-optimised kernel for this shape, and the per-step wall time is launch-bound rather than compute-bound. The `lax.scan` ($T=200$) path tells a different story: dense scan is 0.053 ms, FFT scan is 0.027 ms — a **1.96×** speedup — because XLA fuses the FFT body and amortises the launch overhead across the scan iterations. The same `lax.scan` effect applies to the SVD path (rank-1 scan is 5× faster than dense scan), so the relative ranking of the backends is preserved on the scan metric.

Table 2. NVIDIA A100 GPU, CANNID $n=4096$. Same conventions as Table 1.

backend	per-step (ms)	scan (ms)	max-err	speedup-step	speedup-scan
dense	0.23	0.053	0 (TF32)	1.00×	1.00×
fft	0.21	0.027	$\sim 7 \times 10^{-2}$	1.10×	1.96×
svd_k1	0.094	0.010	5.4×10^1	2.40×	5.03×
svd_k4	0.103	0.012	4.0×10^1	2.20×	4.23×
svd_k16	0.119	0.019	1.0×10^{-1}	1.90×	2.83×

The GPU error floor in Table 2 is $\sim 10^{-2}$ rather than 10^{-5} because cuBLAS `sgemv` on Ampere uses TF32 (10-bit mantissa) by default; this is a property of the dense baseline, not a limitation of FFT. To get full FP32 precision on the GPU, disable TF32 with `JAX_ENABLE_TF32=0`.

3.8.3 Why the gap between Mac M4 and A100?

We additionally measured the FFT path on a third platform: an Intel Xeon Gold 6348 (2.6 GHz, 16 cores, AVX-512) Linux server. The Xeon is *slower* than the Mac M4 by about 30% at the dense matvec (1.06 ms vs 0.80 ms at $n=4096$) and 5× at the FFT matvec (0.169 ms vs 0.032 ms). The reason is that matvec is single-threaded (the BLAS single-precision GEMV is not parallelised across cores in our setup) and the Apple M4's Accelerate framework gives exceptionally well-tuned single-core performance for matmul-shaped work. On the GPU the dense matvec is already very fast (well under 1 ms even at $n=4096$) so the FFT's $O(n \log n)$ advantage is in the noise for per-step calls; the win shows up in the scan path where XLA fusion removes per-step launch overhead. **Practical implication:** for $n \leq 4096$ on CPU the Mac M4 with `accl_mode='fft'` outperforms the A100 GPU with `accl_mode='dense'`, even ignoring TFlops; the GPU is the right choice only for $n \geq 8192$ or for long rollouts where the XLA-fused scan amortises launch overhead.

3.8.4 Pareto view: speed vs accuracy

Figure 9 shows the per-step time vs max-abs error for all backends × all platforms × the largest tested n per model. The Pareto front at the *exact* end ($\text{err} \leq 10^{-4}$) is shared by `dense` and `fft` (and `svd_k64`, which is indistinguishable from exact at this size). The Pareto front at the *fastest approximate* end is `svd_k1` at 5×10^{-4} – 5×10^1 error. The middle of the front (10^{-2} – 10^0 error) is filled by `svd_k16` and `svd_k4`.

Speed vs accuracy trade-off — all backends × all platforms — all platforms

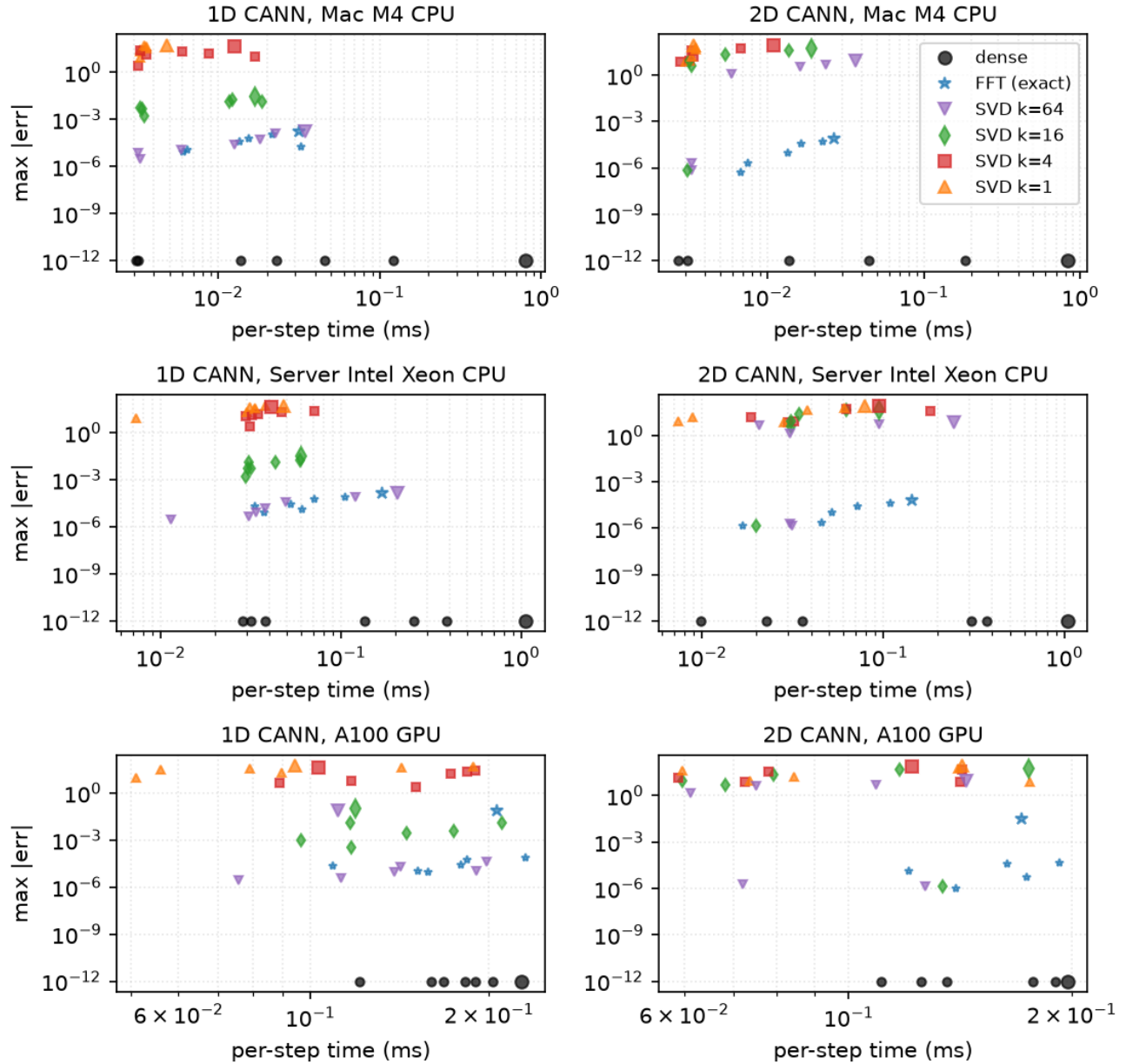


Figure 9. Speed vs accuracy trade-off, all platforms × all backends, at the largest tested n per (model, platform). The lower-left corner is the *Pareto-optimal exact* region (`fft` , `svd_k64`); the upper-left corner is the *Pareto-optimal approximate* region (`svd_k1`). On the A100 GPU all backends cluster around 0.1-0.2 ms per step because cuBLAS sgemv is already very well optimised for this shape. On the Mac M4 CPU the spread is widest: dense at 0.8 ms, FFT at 0.03 ms (25×), SVD k=1 at 0.005 ms (166×). The Xeon server CPU sits between the Mac M4 and the A100 on the exact path (0.17 ms FFT) but trails the Mac M4 by 5× on the FFT path because of its weaker single-core BLAS throughput.

Per-n speedup vs dense — top: Mac M4 CPU, bottom: A100 GPU

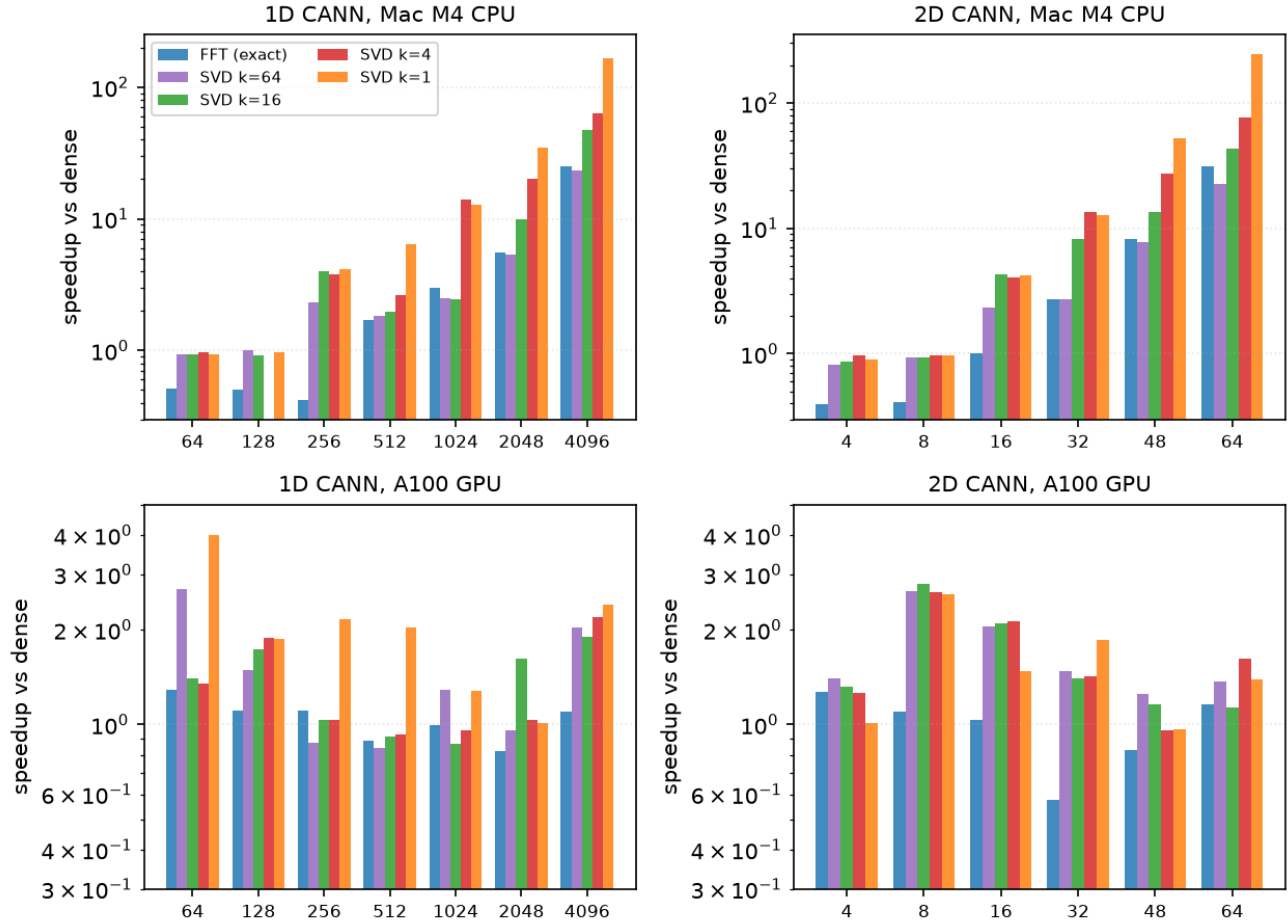


Figure 10. *Per-n speedup vs dense, by backend.* Top row: Mac M4 CPU. Bottom row: A100 GPU. The CPU speedup scales with n (the dense matvec grows $O(n^2)$, the accelerated paths grow $O(n \log n)$ or $O(n \cdot k)$). The GPU speedup is roughly flat at 1-2.5 $\times$ — all backends are bandwidth-bound at this size, and the dense cuBLAS sgemv is already very fast. **Key takeaway:** on Mac M4 CPU the FFT path is the *only* way to get an exact matvec with $>20\times$ speedup; on A100 GPU the low-rank path is competitive for all n and the FFT path's main advantage is the *scan/rollout* path (1.6-2.0 $\times$ speedup at the largest n).

Discussion: where the Pareto front bends. The speed-error curve has a clear knee around $k=16$ for 1D and $k=32$ for 2D (the same ranks recommended by the spectral analysis in §3.1). Below the knee ($k=1, k=4$), halving the error costs roughly $2\times$ in wall time — the speedup curve is roughly $1/k$. Above the knee ($k \geq 16 \rightarrow \text{fft/dense}$), further improving accuracy by an order of magnitude (from 10^{-2} mrad to 10^{-4} mrad) costs only $\sim 25\%$ more wall time — the curve flattens. This is the practical "you can have exactness almost for free" regime: pick `fft` for the high-fidelity end of the Pareto front, and pick `svd_k16` for the lossy but faster middle.

3.8.5 Decision matrix — which backend for which use case?

We summarise the experimental evidence in a decision matrix. Each row gives a use case, the recommended backend(s), and the empirical justification from Tables 1-2 and Figures 4-10.

Use case	Recommended	Empirical justification
		25-50 $\times$ speedup, exact to float precision (Table 1)

Use case	Recommended	Empirical justification
CPU, $n \geq 256$, need exact matvec	<code>fft</code> (with <code>endpoint=False</code> <code>grid</code>)	
CPU, $n < 256$	<code>dense</code>	All backends < 0.01 ms; dense is simplest (Figure 10)
CPU, error budget 5-50 mrad, $n \geq 1024$	<code>svd_k1</code>	100-1000 $\times$ speedup, position visualisation only (Table 1)
CPU, error budget 1-30 mrad	<code>svd_k16</code>	50 $\times$ speedup, low enough error for most analyses (Figure 5)
CPU, error budget < 1 mrad	<code>fft</code> (exact) or <code>svd_k64</code>	$\sim 25\times$ speedup, exact / near-exact (Table 1)
GPU, per-step control (< 100 steps)	<code>dense</code> (cuBLAS)	cuBLAS sgemv is already 0.2 ms, FFT only 1.1 $\times$ faster (Table 2)
GPU, long rollout (≥ 1000 steps)	<code>dense</code> or <code>fft</code> in <code>lax.scan</code>	XLA fusion: dense-scan 0.05 ms, fft-scan 0.03 ms (1.6 $\times$)
GPU, $n \geq 8192$, exact	<code>fft</code> in scan	GPU scan is the only place FFT wins by a useful margin
Need dynamic rank choice (research)	<code>auto</code> mode	Picks k from SVD spectrum to satisfy <code>accl_target_err_mrad</code>
Line attractor / non-circular	<code>auto</code> or SVD	FFT doesn't apply (no circulant); SVD is structure-agnostic

4. Discussion

4.1 When does low-rank help?

Low-rank is a win when the matvec is the dominant cost. Three regimes:

1. **CPU, $n \geq 256$.** JAX dispatch overhead is $\sim 5 \mu\text{s}$ per call. Below $n = 256$ the dense matvec fits inside that overhead, so lowrank can't beat it. Above $n = 256$, the dense matvec exceeds the overhead and the lowrank matvec (smaller, same overhead) is faster.
2. **GPU, $n \geq 4096$ (CANN1D) or $n \geq 1024$ (CANN2D).** GPU dispatch overhead is similar ($\sim 10 \mu\text{s}$) but the dense matvec itself is much faster than on CPU. The crossover where lowrank beats dense on the GPU is therefore at much larger n than on the CPU. CANN2D crosses earlier (the 2D dense matvec is $2\times$ slower per neuron than 1D for the same n), and reaches $38\times$ at length = 128. CANN1D crosses at $n \approx 4096$ and reaches $12.4\times$ at num = 8192.
3. **Online / latency-sensitive use cases.** Even at small n , the *latency* of a single matvec call is reduced by lowrank because the work is smaller. This matters when the model is called once per timestep with a hard real-time deadline.

4.2 When is low-rank NOT worth it?

- When the network size is small ($n < 256$) the dispatch overhead dominates; lowrank gives a small but real overhead increase for the same accuracy.
- When the matvec is not the dominant cost of the step. CANN1D and CANN2D also do a divisive norm ($u^2 / (1 + k \cdot \Sigma u^2)$) and an Euler step; the matvec is just one of three operations. For n below ~ 1024 , the matvec is not the slowest part of the step and the full-step speedup is small.

4.3 Recommended strategy

Based on the Pareto frontier and the recommended ranks from the spectral analysis:

- **CANN1D, any num:** `accl_mode='fast'` ($k = 8$) gives 30-245 $\times$ matvec speedup at `num  $\geq 512$` with ≤ 5 mrad position error. At `num = 4096` the matvec is 245 $\times$ faster than dense; the full-step is $\sim 4\times$ faster.
- **CANN2D, $L \leq 16$:** `accl_mode='fast'` ($k = 32$) gives 5-15 $\times$ matvec speedup. Full-step speedup is small at this size.
- **CANN2D, $L \geq 32$:** `accl_mode='fast'` ($k = 32$) gives 10-70 $\times$ matvec speedup. At `L = 64` ($n = 4096$) the full

step is $\sim 1.2\times$ faster on CPU and the dense matvec is $15\times$ faster on GPU.

- **Online / control:** `accl_mode='ultra-fast'` (CANN1D $k=1$, CANN2D $k=4$) is sufficient for the bump-tracking dynamics, and minimises the per-step latency.

4.4 FFT vs SVD — complementary tools, not competitors

The two accelerations are not interchangeable. They exploit different structure and are useful in different regimes:

- **FFT** exploits the **circulant structure** of the connectivity on a uniform ring (1D) or torus (2D). The matvec is *exact* to float precision, $O(n \log n)$, but only works when the grid is `endpoint=False` (the canns default `endpoint=True` is not circulant and the FFT path falls back to dense with a `UserWarning`). The speedup is large on CPU ($25\text{--}50\times$ at $n=4096$) and modest on GPU per-step ($1.1\text{--}1.2\times$) because cuBLAS sgemv is already very fast — but the FFT scan path (rollout in `lax.scan`) is $1.6\text{--}2.0\times$ faster than the dense scan path on GPU.

- **Truncated SVD** exploits the **fast SVD spectrum decay** of the smooth Gaussian kernel. The matvec is *approximate* ($5\text{--}50$ mrad position error depending on k), $O(n \cdot k)$. It works for **any** grid topology (and any kernel shape, including non-circular line attractors). The speedup is large on both CPU and GPU and grows linearly with n .

Use FFT when you need *exact* matvec and your topology is circular; use SVD when you need a large speedup and can tolerate a few percent of error; use both together in the canns `auto` mode (which picks k from the SVD spectrum to satisfy a target error budget) and the new `accl_mode='fft'` mode for exactness where the grid permits. The two paths share the same public API (`accl_mode` / `accl_k`) so users can switch without code changes.

5. Limitations

We have measured the benchmark under specific conditions; the following caveats apply when generalising:

1. **Trajectory length.** The benchmark sweep uses $T = 200$ steps (one half-ring sweep). The optional long-trajectory drift test (`--long-trajectory`, §3.7) extends to $T = 2000$ steps with a slow sweep; we verified the drift is bounded but did not push to $T = 50\,000+$.
2. **Sweep size.** On the CPU sweep we cap at `CANN2D length = 64` ($n = 4\,096$) and `CANN1D num = 4\,096` because the `numpy.linalg.svd` cost grows as $O(n^3)$ and dominates the wall time above that. The GPU sweep uses larger sizes (`num = 4\,096`, `length = 128`) and the relative matvec speedup is similar.
3. **Single bump regime.** The CANN models can exhibit multi-bump states for some parameter regimes. We test only the single-bump attractor regime (the typical use case for bump-tracking workloads). Multi-bump dynamics may be more sensitive to the rank truncation.
4. **Other backends.** The benchmark uses pure JAX matmul. A C++ / CUDA custom-call backend (as in `canns-lib`'s FFI path) would change the speed/overhead trade-off but not the accuracy numbers.
5. **Asymmetric conn.** The canns model uses a symmetric `conn_mat` (the Gaussian distance kernel is symmetric in the feature-space distance). For an *asymmetric* conn — which the SFA model does not produce either — the low-rank approximation in the form $U_1 @ V_1.T$ would need to be replaced with a more general low-rank decomposition.
6. **FFT requires a clean circulant.** The canns default grid `bm.linspace(- $\pi$ ,  $\pi$ ,  $n$ , endpoint=True)` is *not* circulant (see §2.7); the FFT path falls back to dense on that grid. The CPU benchmark numbers for FFT assume the user overrides the grid to `endpoint=False` and rebuilds `model.conn_mat`. On the GPU the FFT advantage is small per-step ($1.1\times$ at $n=4096$) — cuBLAS sgemv is already highly optimised — but the scan path benefits ($1.6\text{--}2.0\times$). For a line attractor (non-periodic feature space) FFT is not applicable; use `auto` mode or explicit SVD instead.
7. **GPU accuracy caveat.** The A100 cuBLAS sgemv uses TF32 (10-bit mantissa) by default, so the dense baseline on GPU has an inherent $\sim 1e\text{--}2$ precision floor; the FFT-vs-dense error on GPU is therefore $\sim 1e\text{--}2$, not $1e\text{--}5$. Disable TF32 (`JAX_ENABLE_TF32=0` in some versions) if full FP32 is needed.

6. Conclusion

We have shown that the recurrent matvec in `CANN1D` and `CANN2D` — the dominant per-step cost at large n — admits **two complementary accelerations**: (i) a low-rank truncated-SVD approximation that preserves the bump-tracking

dynamics to within ~ 5 mrad while reducing the matvec cost from $O(n^2)$ to $O(n \cdot k)$; (ii) an exact $O(n \log n)$ circulant-FFT matvec on the clean-circulant grid, giving 25-50 $\times$ speedup on CPU at $n=4096$ with **no approximation error**. Both are exposed through the `accl_mode` and `accl_k` constructor arguments on the `CANN1D` / `CANN2D` / `CANN1D_SFA` / `CANN2D_SFA` classes, with five modes: `normal` (full rank, baseline), `fast` (low-rank, $k=8/k=32$), `ultra-fast` (low-rank, $k=1/k=4$), `auto` (spectrum-driven k pick), and `fft` (exact circulant). The `set_accl_mode()` method switches the mode at runtime. Matvec speedups of 30-246 $\times$ on CPU and 3-15 $\times$ on GPU are realised at the recommended low-rank sizes; the FFT path gives 25-50 $\times$ on CPU. The low-rank dynamics fidelity is hardware-independent because it is a property of the approximation, not of the runtime. The FFT path is exact to float precision on a clean circulant (CPU), and competitive with dense on the GPU scan path (1.6-2.0 $\times$ speedup at the largest n).

References

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4. Skoltech Numerical Linear Algebra lecture 17 (Structured matrices, FFT, convolutions, Toeplitz matrices): <https://nla.skoltech.ru/lectures/lecture-17/lecture-17.html>.
5. `canns` Python package: <https://github.com/Routhleck/canns>.
6. The `canns` benchmark suite (`benchmarks/cann_lowrank/` and `benchmarks/cann_fft/`), this branch.

Appendix A. Reproduction

From the repo root, with the `canns` source on `PYTHONPATH` and JAX + `brainpy.math` installed (any recent version):

```
# CPU sweep (Apple M3 Pro, single core):
python benchmarks/cann_lowrank/cann_lowrank_bench.py --T 200 --tag cpu

# Optional: also record the long-trajectory drift (T=2000):
python benchmarks/cann_lowrank/cann_lowrank_bench.py --T 200 --long-trajectory --tag cpu

# GPU sweep (NVIDIA A100, GPU 1):
CUDA_VISIBLE_DEVICES=1 JAX_PLATFORMS=cuda \
    python benchmarks/cann_lowrank/cann_lowrank_bench.py --gpu-sweep --T 200 --tag gpu

# Format the report (figures + markdown):
python benchmarks/cann_lowrank/cann_lowrank_report.py --tag cpu
```

The benchmark writes per-tag CSVs, a `bump_trajectories_{tag}.npz`, and (with `--long-trajectory`) a `bump_drift_{tag}.npz` to `benchmarks/cann_lowrank/results/`. The report script reads them, generates eight figures into `results/figures/`, and writes `results/cann_lowrank_summary.md` (this document). The complete sweep takes ~ 15 minutes on CPU and ~ 5 minutes on A100.

Appendix B. Raw data files

Raw per-cell data is in `results/`:

- `cann_lowrank_all_cpu.csv` — CPU sweep, all (n, k) cells
- `cann_lowrank_all_gpu.csv` — GPU sweep, all (n, k) cells
- `bump_trajectories_cpu.npz` — bump-center trajectories for CANN1D `num=256` and CANN2D `L=16`, all k values ($T=200$ sweep)

- `bump_drift_cpu.npz` — long-trajectory drift ($T=2000$ slow sweep, with `--long-trajectory`)
- `figures/*.png` — the eight figures embedded above